

Microbial Growth and Physiology

Microbiology - LG-1

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Growth, Survival and Death

Microbial population is constantly balanced by:

- Adaptation to the environment
- Competition with other organisms

Bacterial physiology is the study of their:

- Growth
- Survival
- Evolution



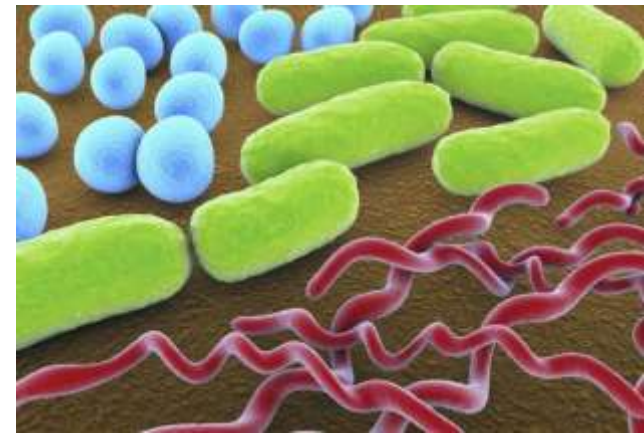
The Growth

Microbial growth:

- An increase in the number of unicellular organisms
- The growth is referred to as culture

The growth can be measured in two ways:

- **Microbial concentration:** The number of viable cells per unit volume of culture
- **Biomass density:** Dry weight of cells per unit volume of culture



Growth curve phases

1st: Lag phase

- Bacterial cells adapt to their new environment
- Enzymes and intermediates are formed and accumulate to start the growth.

2nd: Exponential phase:

- Constant synthesis of cell materials
- Exponential increase in mass

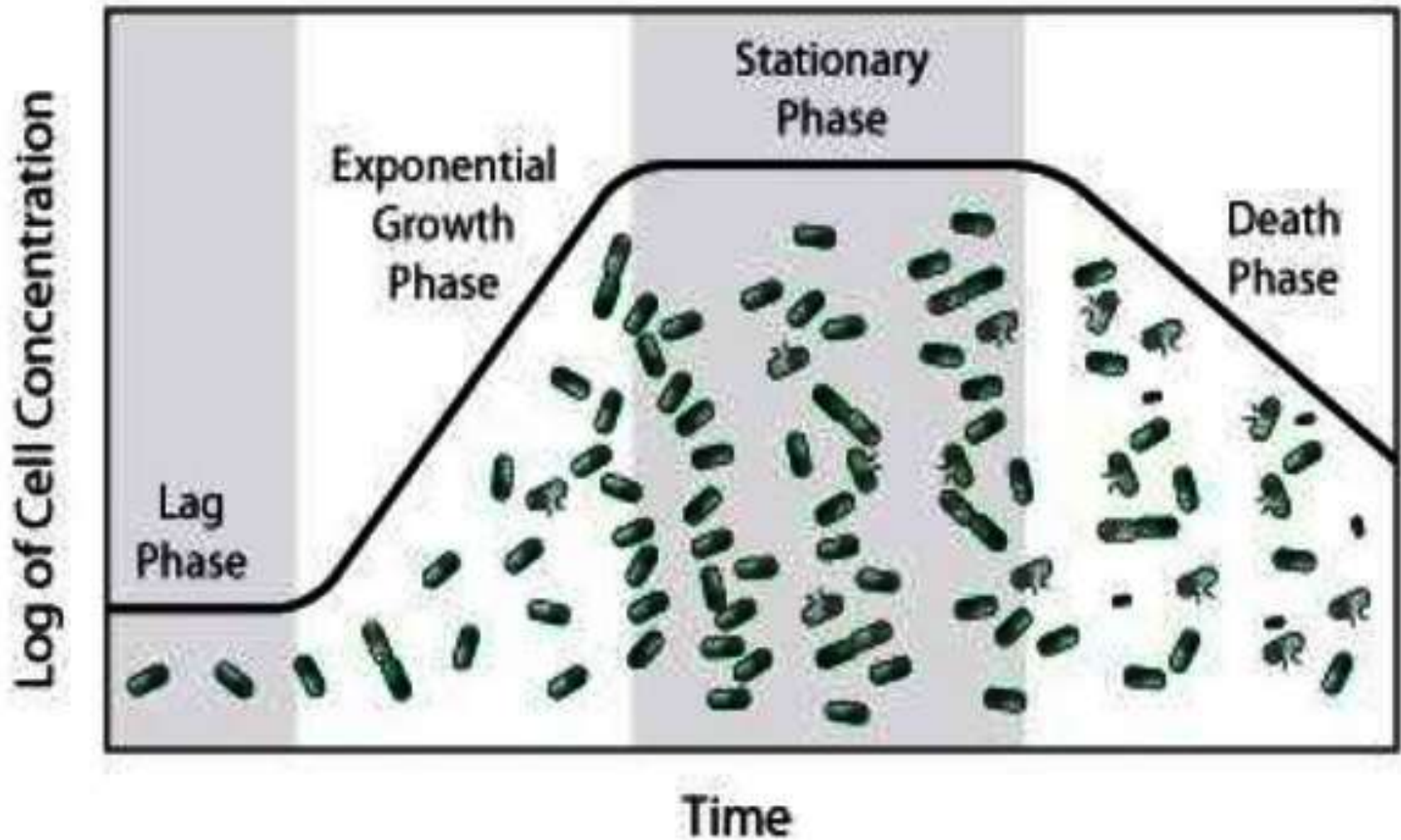
3rd: Stationary phase:

- A balanced rate of growth to death due to:
 - Exhaustion of nutrients
 - Accumulation of toxic products
- Total cell count increases but viable count stays constant

4th: Decline (Death) phase:

- Increased death rate
- A small number may persist for months or years

The Growth Curve



Bacterial growth outcomes

Dormancy: bacterial species that cause chronic infections.

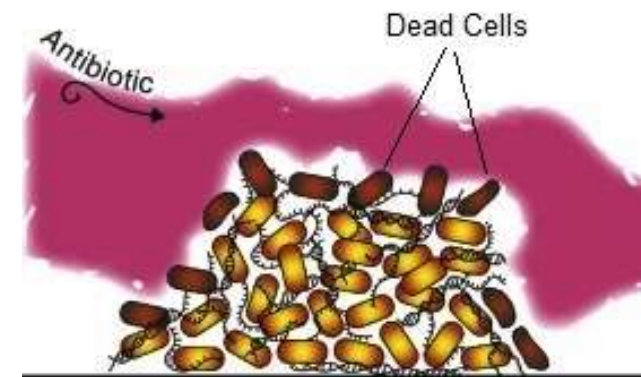
Spore formation: Protecting bacteria from dryness and lack of nutrients (e.g. *Bacillus anthracis*)

VBNC (Viable but non-culturable):

- A genetic response to environmental stress (e.g. starving)
- The growth can be resumed when the stress is removed

Biofilm:

- Layers of multiple populations of bacteria at different growth phases.
- Medically important due to:
 - Immune evasion
 - Shielding from antibiotic effects

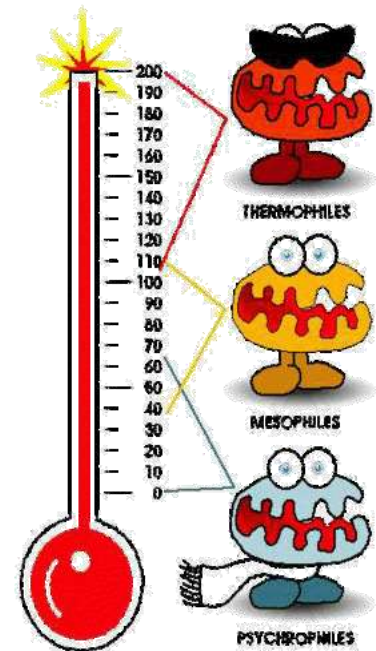


The Growth Physiology

Temperature:

- **Psychrophilic** (cold loving): grow below 20°C (e.g. *Flavobacterium spp*)
- **Mesophilic**: grow between 20°C and 45°C
- **Thermophilic**: grow at above 45°C

The optimum temperature for most pathogenic bacteria is 37°C.



The Growth Physiology

Atmosphere:

Depending on the oxygen requirement, bacteria are divided into:

Strict (Obligate) Aerobes: require O_2 for growth

e.g. *Pseudomonas aeruginosa*

Microaerophilic: grow best in the presence of low oxygen levels

e.g. *Helicobacter pylori*

Anaerobes: incapable of growth on exposure to O_2

e.g. *Clostridium tetani*

Facultative anaerobe: aerobes but also grow in the absence of O_2

e.g. *Staphylococcus aureus*

Capnophilic: require high CO_2 levels

e.g. *Neisseria meningitidis*

The Growth Physiology

pH:

The optimum pH for majority of pathogenic bacteria is neutral or slightly alkaline (7.2 – 7.6).

Lactobacilli grow best under acidic pH whereas ***Vibrio cholera*** requires alkaline pH

Water:

- ***Treponemal pallidum*** is highly sensitive to drying
- ***Staphylococci spp*** can tolerate months without water.
- Spores are resistant to dryness (may survive for several decades)

Osmotic effects:

Bacteria with a proper cell wall are more tolerant to osmotic variation due to the mechanical strength of their cell walls

References

- Jawetz, Melnick & Adelberg' Medical Microbiology, 28th Edition (2019)
- Levinson W. Review of Medical Microbiology and Immunology, 13th Edition (2014)



Scientists discover a new superbug.